Reducing Friction

Aim: To reduce the force of friction between a lego car and a block of wood.

Hypothesis:

Which material will have the most friction: _____

Which material will have the least friction: _____

Will you expect the same results for sandpaper for both sides of your lego car? Why? _____

Results:

Surface	Angle (◦)	
	Lego Side	Wheel side
Wood	15◦	2◦
Grip mat	35◦	5◦
Sandpaper	60◦	7◦
Material	35◦	4◦
Carpet	25◦	8◦
Wax paper	10◦	2◦
Transparent sheet	23◦	3◦

Draw a bar graph showing the results from the table above.

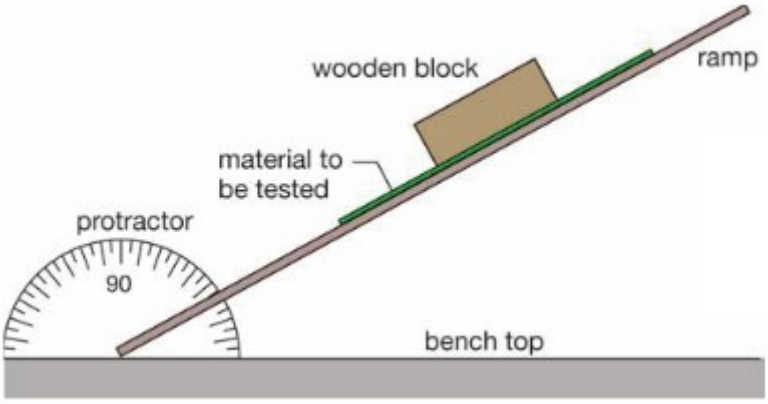
Discussion:

1. List the following variables:
   (a) Controlled

   (b) Independent

   (c) Dependent

2. Draw arrows and labels of the forces acting on the lego car



3. Which surface had the most friction acting on the lego car? Explain why.
4. Which surface had the least amount of friction acting on the lego car? Explain why?
5. Why were your results different for lego side and wheel side of the car? Which side has the least amount of friction?

Conclusion: Summary paragraph relating friction to surface